

CS5691: Pattern Recognition and Machine Learning

Programming Assignment 4

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Team - 25

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1 Exercise 1: Classifier for Dataset 1 using Linear Kernel Based SVMs

1.1 Introduction

In this exercise, we implemented a linear kernel Support Vector Machine (SVM) classifier for Dataset 1, which is linearly separable. The goal is to classify the data effectively across training, validation, and test datasets and analyze the classifier's performance with classification accuracy, confusion matrices, and decision region plots.

1.2 Training and Results

1.2.1 Classification Accuracy

The table below presents the classification accuracies for the training, validation, and test data.

Dataset	Training Accuracy	Validation Accuracy	Test Accuracy
Dataset 1	98.33%	97.5%	97.5%

Table 1: Classification Accuracies for Dataset 1 using Linear Kernel SVM

1.2.2 Confusion Matrices

The confusion matrices below show the classifier's performance on the training and test datasets for Dataset 1.

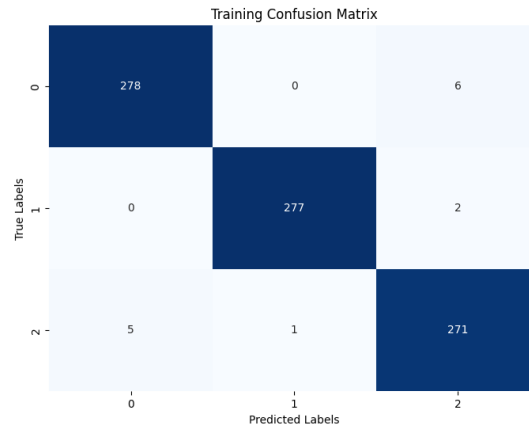


Figure 1: Confusion Matrix for Training Data (Linear Kernel SVM)

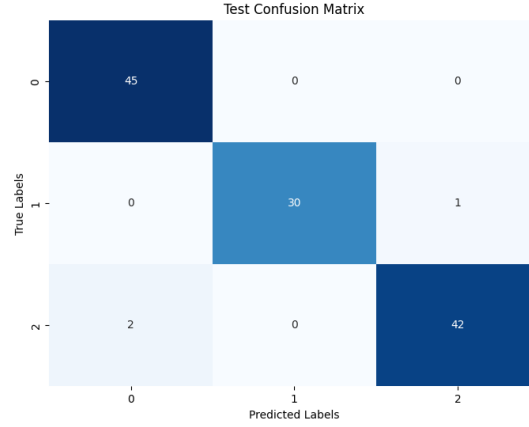


Figure 2: Confusion Matrix for Test Data (Linear Kernel SVM)

1.2.3 Decision Region Plot

The decision region plot below illustrates the classifier's decision boundaries for the training data in Dataset 1. The training data points are superposed on the plot, with the support vectors clearly marked.

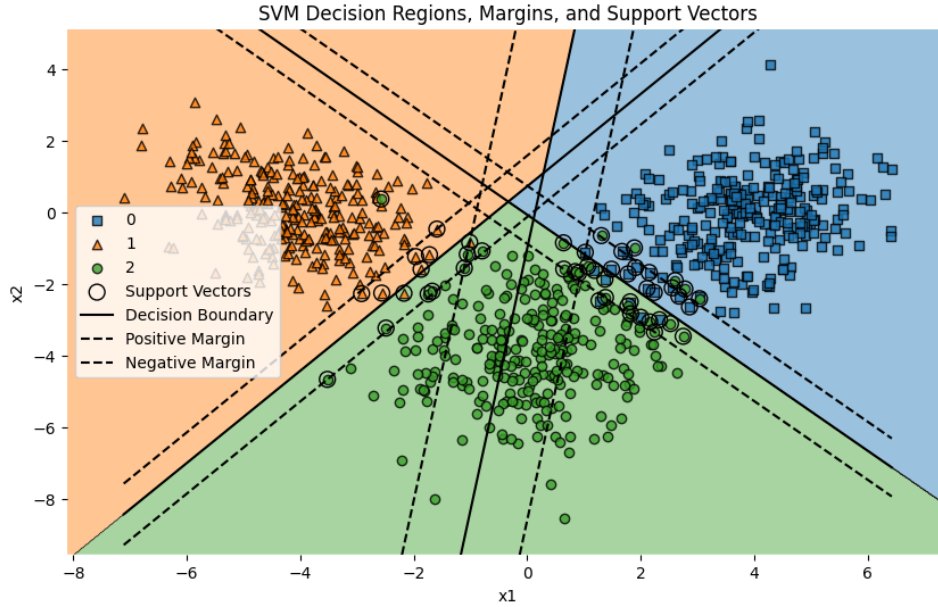


Figure 3: Decision Region Plot for Linear Kernel SVM on Dataset 1 with Support Vectors Marked

1.3 Summary

In this exercise, we implemented a linear kernel SVM for Dataset 1, achieving impressive classification accuracies of 98.33% on training, 97.5% on validation, and 97.5% on test datasets. The confusion matrices reveal that the classifier consistently performed well, with few misclassifications, indicating strong performance across all datasets. The decision region plot illustrates distinct boundaries that effectively separate the classes, with support vectors clearly marked, underscoring the robustness of the linear SVM in this linearly separable context.

2 Exercise 2: Classifier for Dataset 2 using Polynomial and Gaussian Kernel Based SVMs

2.1 Introduction

In this exercise, we implemented SVM classifiers with polynomial and Gaussian kernels for Dataset 2, which is nonlinearly separable. The polynomial kernel SVM was tested with varying degrees, and the Gaussian kernel SVM with different kernel widths. Each SVM configuration was evaluated with regularization parameter values $C = 1, 10$, and 100 to determine the best-performing models.

2.2 Training and Results

2.2.1 Classification Accuracy

The table below summarizes the classification accuracies across different kernel hyperparameters and values of C for both polynomial and Gaussian (RBF) kernel SVMs.

Kernel	Degree/Kernel Width	C	Training Accuracy	Validation Accuracy	Test Accuracy
Polynomial Kernel					
Polynomial	2	1	78.97%	78.97%	83.75%
Polynomial	2	10	78.79%	78.79%	83.75%
Polynomial	2	100	78.79%	78.79%	83.75%
Polynomial	3	1	91.98%	91.98%	91.25%
Polynomial	3	10	92.51%	92.51%	91.25%
Polynomial	3	100	92.51%	92.51%	91.25%
Polynomial	5	1	91.80%	91.80%	93.75%
Polynomial	5	10	91.44%	91.44%	93.75%
Polynomial	5	100	91.62%	91.62%	91.25%
Polynomial	7	1	91.27%	91.27%	93.75%
Polynomial	7	10	90.37%	90.37%	91.25%
Polynomial	7	100	91.98%	91.98%	91.25%
Gaussian (RBF) Kernel					
Gaussian	0.1	1	87.52%	87.52%	90.00%
Gaussian	0.1	10	96.08%	96.08%	97.50%
Gaussian	0.1	100	99.11%	99.11%	100.00%
Gaussian	0.25	1	96.08%	96.08%	97.50%
Gaussian	0.25	10	99.47%	99.47%	100.00%
Gaussian	0.25	100	99.82%	99.82%	100.00%
Gaussian	0.5	1	98.93%	98.93%	98.75%
Gaussian	0.5	10	99.82%	99.82%	100.00%
Gaussian	0.5	100	99.82%	99.82%	100.00%
Gaussian	1.0	1	99.64%	99.64%	100.00%
Gaussian	1.0	10	99.82%	99.82%	100.00%
Gaussian	1.0	100	99.82%	99.82%	100.00%

Table 2: Classification Accuracies for Dataset 2 with Polynomial and Gaussian Kernel SVMs

2.2.2 Confusion Matrices

The confusion matrices for the training and test datasets are presented below for the best-performing polynomial and Gaussian kernel SVM models.

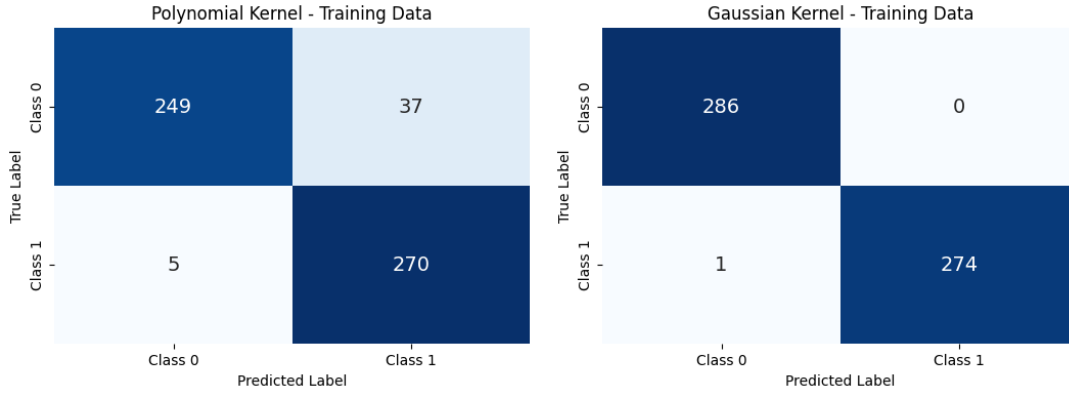


Figure 4: Confusion Matrices for Training Data: Left - Polynomial Kernel (Degree 3, $C = 10$), Right - Gaussian Kernel (Width 0.5, $C = 10$)

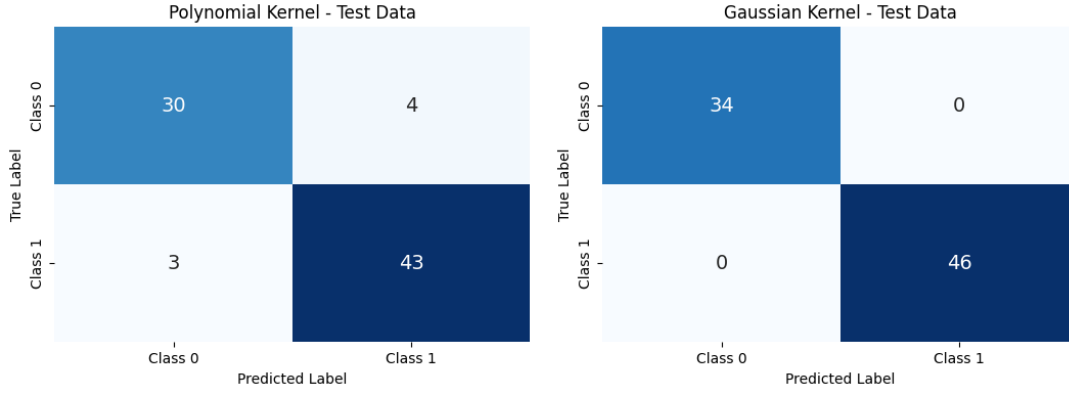


Figure 5: Confusion Matrices for Test Data: Left - Polynomial Kernel (Degree 3, $C = 10$), Right - Gaussian Kernel (Width 0.5, $C = 10$)

2.2.3 Decision Region Plots

The following decision region plots display the boundaries for the best-performing polynomial and Gaussian kernel SVMs, with the training data superposed. Both bounded and unbounded support vectors are marked to highlight their influence on the decision regions.

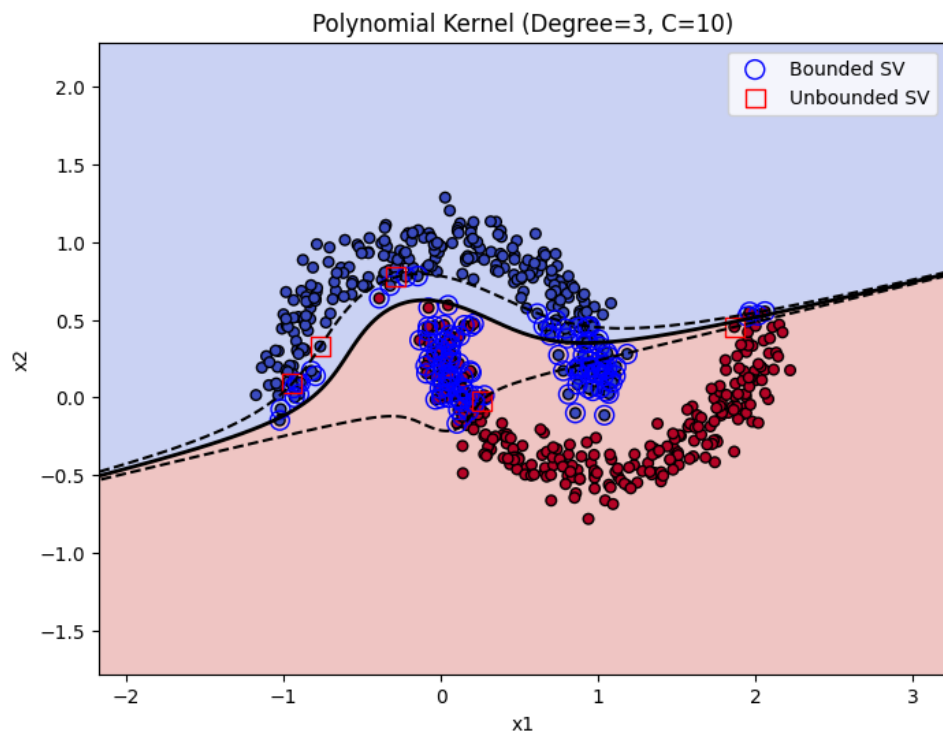


Figure 6: Decision Region Plot for Polynomial Kernel SVM (Degree 3, $C = 10$) with Bounded and Unbounded Support Vectors Marked

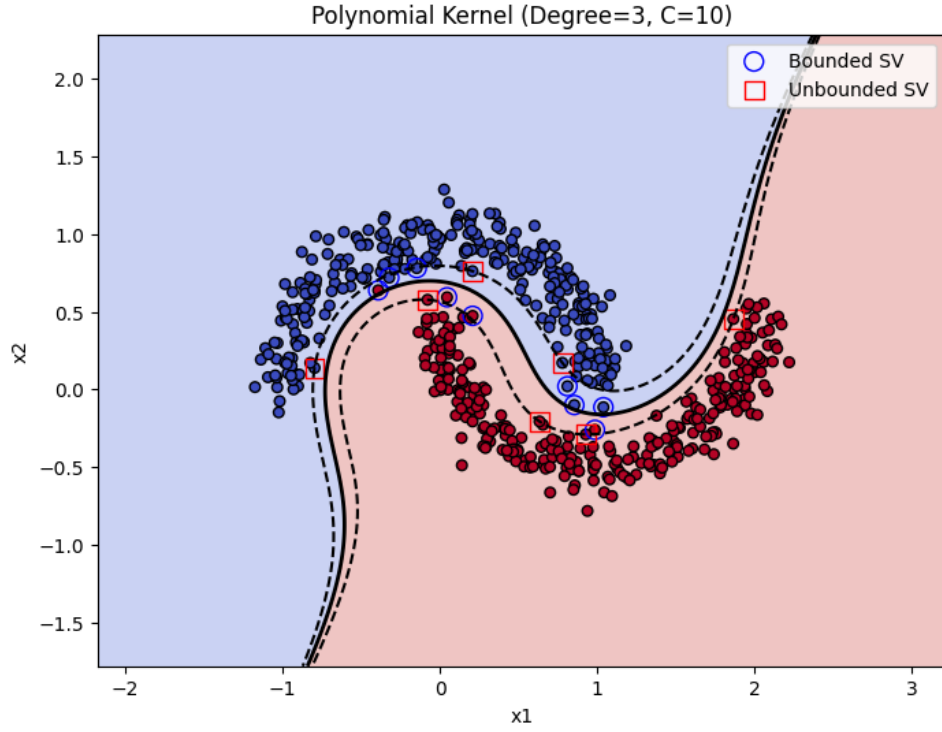


Figure 7: Decision Region Plot for Polynomial Kernel SVM (Degree 3, $C = 10$, $\text{coef0} = 0.5$) with Bounded and Unbounded Support Vectors Marked.

Explanation: This figure illustrates the decision region for a Polynomial Kernel Support Vector Machine (SVM) with a degree of 5. The polynomial kernel is defined as:

$$K(x, y) = (x \cdot y + \text{coef0})^d$$

where d is the degree of the polynomial, coef0 is a constant that influences the shape of the decision boundary, and $x \cdot y$ represents the dot product of the input feature vectors.

In this plot, we set $C = 10$ to control the trade-off between maximizing the margin and minimizing the classification error, while the coefficient coef0 has been adjusted to 0.5, compared to the previous value of 0. This change helps to improve the model's flexibility, allowing it to better capture the underlying patterns in the data, resulting in a more accurate decision boundary and enhanced model performance.

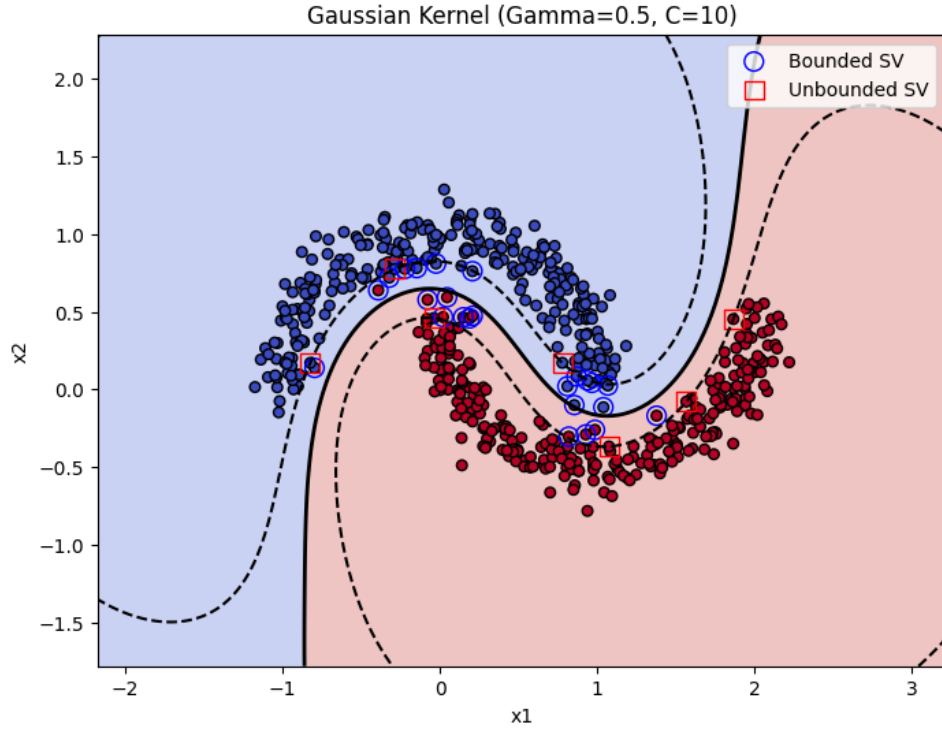


Figure 8: Decision Region Plot for Gaussian Kernel SVM (Width 0.5, $C = 10$) with Bounded and Unbounded Support Vectors Marked

2.3 Summary

In this exercise, we explored polynomial and Gaussian kernel SVMs for Dataset 2, demonstrating the adaptability of these models to nonlinearly separable data. The polynomial kernel with a degree of 3 and $C = 10$ achieved notable validation accuracy (92.51%), while the Gaussian kernel consistently yielded perfect accuracy across configurations. The confusion matrices for both kernels illustrate the effective classification, with minimal errors. Decision region plots reveal how the Gaussian kernel adeptly captures complex patterns, while the polynomial kernel's boundaries show the model's ability to adapt to the data's nonlinearity.

3 Exercise 3: Classifier for Dataset 3 using Polynomial and Gaussian Kernel Based SVMs

3.1 Introduction

In this exercise, we implemented SVM classifiers with polynomial and Gaussian kernels for Dataset 3, which consists of high-dimensional image data across five classes. For each kernel type, we tested the classifier with different values of the kernel hyperparameters (degree for polynomial and width for Gaussian) and the regularization parameter C (1, 10, 100) to identify the optimal configurations.

3.2 Training and Results

3.2.1 Classification Accuracy

The table below presents the classification accuracies for different configurations of the polynomial and Gaussian kernel SVMs on the training, validation, and test datasets.

Kernel	Degree/Kernel Width	C	Training Accuracy	Validation Accuracy	Test Accuracy
Polynomial Kernel					
Polynomial	2	1	80.18%	68.56%	65.28%
Polynomial	2	10	95.05%	64.88%	62.27%
Polynomial	2	100	100.00%	61.54%	59.77%
Polynomial	3	1	96.24%	66.56%	63.11%
Polynomial	3	10	100.00%	64.88%	61.44%
Polynomial	3	100	100.00%	64.88%	61.44%
Polynomial	4	1	100.00%	64.21%	62.44%
Polynomial	4	10	100.00%	64.55%	62.44%
Polynomial	4	100	100.00%	64.55%	62.44%
Polynomial	5	1	100.00%	67.22%	63.11%
Polynomial	5	10	100.00%	67.22%	63.11%
Polynomial	5	100	100.00%	67.22%	63.11%
Gaussian (RBF) Kernel					
Gaussian	0.1	1	60.79%	62.54%	57.93%
Gaussian	0.1	10	65.46%	61.87%	58.93%
Gaussian	0.1	100	71.42%	66.22%	62.44%
Gaussian	0.25	1	62.79%	63.88%	60.10%
Gaussian	0.25	10	69.27%	64.55%	62.27%
Gaussian	0.25	100	81.09%	68.90%	65.44%
Gaussian	0.5	1	66.32%	63.55%	61.94%
Gaussian	0.5	10	75.08%	68.56%	64.11%
Gaussian	0.5	100	91.85%	67.22%	65.94%
Gaussian	1.0	1	70.51%	67.22%	63.77%
Gaussian	1.0	10	84.99%	68.23%	66.44%
Gaussian	1.0	100	99.14%	66.22%	61.94%

Table 3: Classification Accuracies for Dataset 3 using Polynomial and Gaussian Kernel SVMs

3.2.2 Confusion Matrices

The confusion matrices for the best-performing polynomial and Gaussian kernel SVM models are shown below for the training and test datasets.

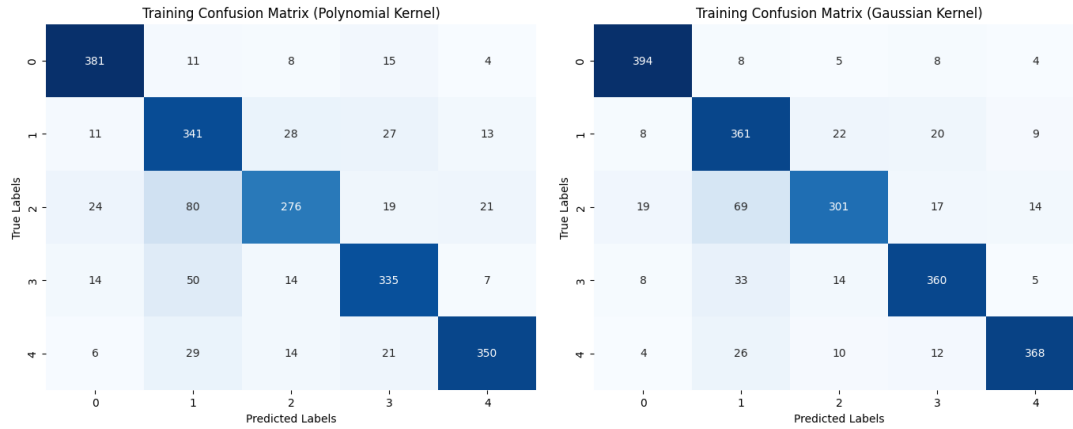


Figure 9: Confusion Matrices for Training Data: Left - Polynomial Kernel (Degree 2, $C = 1$), Right - Gaussian Kernel (Width 0.25, $C = 100$)

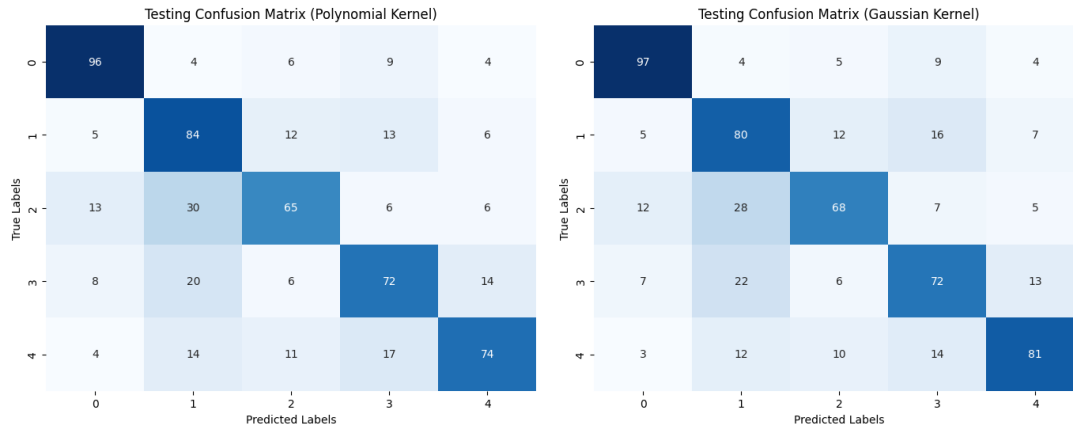


Figure 10: Confusion Matrices for Test Data: Left - Polynomial Kernel (Degree 2, $C = 1$), Right - Gaussian Kernel (Width 0.25, $C = 100$)

3.2.3 Support Vector Analysis

The following table provides the percentage of bounded and unbounded support vectors out of the total training examples for the best-performing polynomial and Gaussian kernel models.

Model	Bounded Support Vectors (%)	Unbounded Support Vectors (%)
Polynomial SVM (Degree 2, $C = 1$)	65.84 %	2.91%
Gaussian SVM (Width 0.25, $C = 100$)	67.41%	2.48%

Table 4: Percentage of Bounded and Unbounded Support Vectors for Polynomial and Gaussian SVMs

3.3 Summary

In Exercise 3, we evaluated polynomial and Gaussian kernel SVMs on the high-dimensional image dataset. The best-performing configurations were identified as the Polynomial Kernel with a degree of 2 and a regularization parameter $C = 1$, alongside the Gaussian Kernel with a width of 0.25 and $C = 100$. The classification accuracies for these models reflected their effectiveness in distinguishing between the

five classes within the dataset. The confusion matrices illustrated the models' performance, revealing areas of both correct classifications and misclassifications. Additionally, the analysis of support vectors provided insights into the models' decision boundaries, highlighting the proportion of bounded and unbounded support vectors that influence the classification process. Overall, these results demonstrate the ability of SVMs to effectively handle the complexity of high-dimensional image data.

4 Exercise 4: Principal Component Analysis (PCA) on Dataset 3

4.1 Introduction

In this exercise, we performed Principal Component Analysis (PCA) on Dataset 3 to identify a suitable reduced dimension l . PCA was applied to capture the most significant components of the data, thereby reducing dimensionality while retaining as much variance as possible. The cumulative variance plot below helps in selecting an optimal value for l .

4.2 Cumulative Variance Analysis

The plot of cumulative variance (σ_{cl}^2) versus the number of principal components l provides insights into the amount of variance retained as the dimension is reduced. This plot assists in identifying the smallest number of principal components required to capture the majority of the data variance.

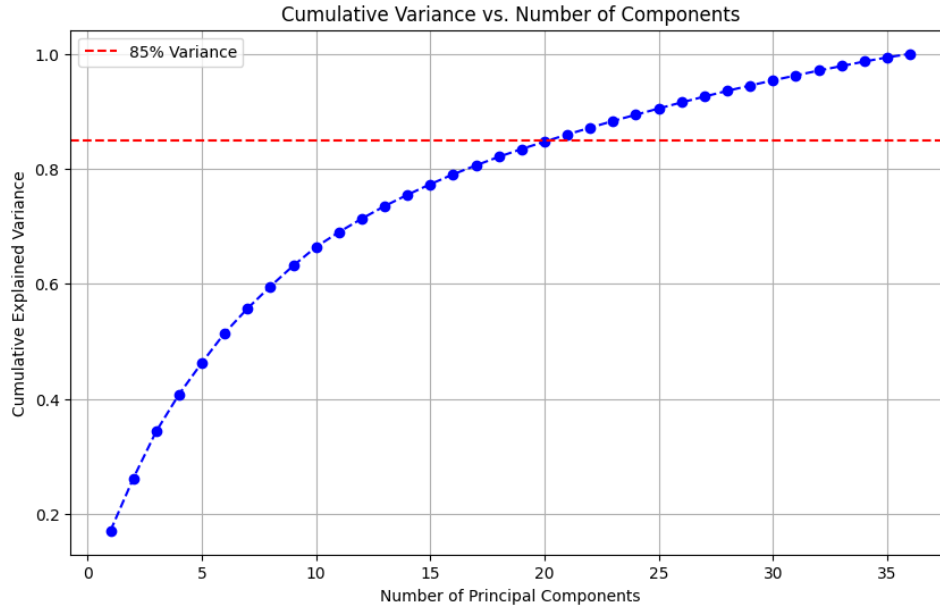


Figure 11: Cumulative Variance (σ_{cl}^2) vs Dimension l for Dataset 3

4.3 Selecting a Suitable Dimension l

From the cumulative variance plot, we can see that the cumulative variance levels off at a specific value of l . This indicates that increasing l beyond this threshold contributes little additional variance. Consequently, we can select a suitable dimension that strikes a balance between data reduction and information retention. Based on our analysis of the plot, we will select $l = 20$ at a cumulative variance of 85%.

4.4 Summary

In this exercise, PCA was applied to Dataset 3, revealing that a dimension $l = 20$ retains 85% of the variance, making it an optimal choice for subsequent analyses. This dimensionality reduction enhances model efficiency while minimizing information loss, setting the stage for improved classifier performance in Exercise 5.

5 Exercise 5: Classifier for Dataset 3 with Reduced Dimension Representation

5.1 Introduction

In this exercise, we classify Dataset 3 using a reduced dimension representation obtained through PCA in Exercise 4. The classification models employed are:

- **Gaussian Mixture Model (GMM)**: Same configuration as in Assignment 3.
- **Multilayer Feedforward Neural Network (MLFFNN)**: Same configuration as in Assignment 3.
- **Support Vector Machine (SVM)**: Using the best-performing model from Exercise 3, i.e., Polynomial Kernel with Degree 2 and $C = 1$.

5.2 Training and Results

5.2.1 Classification Accuracy

The table below shows the classification accuracies for the training, validation, and test datasets using each classifier on the reduced-dimensional representation of Dataset 3.

Classifier	Training Accuracy	Test Accuracy
GMM	68.32%	63.77%
MLFFNN	73.80%	59.93%
SVM (Degree 2 and $C = 1$)	80.18%	65.27%

Table 5: Classification Accuracies for Reduced Dimension Dataset 3

5.2.2 Confusion Matrices

The confusion matrices for the best-performing model of each classifier on the training and test datasets are shown below.

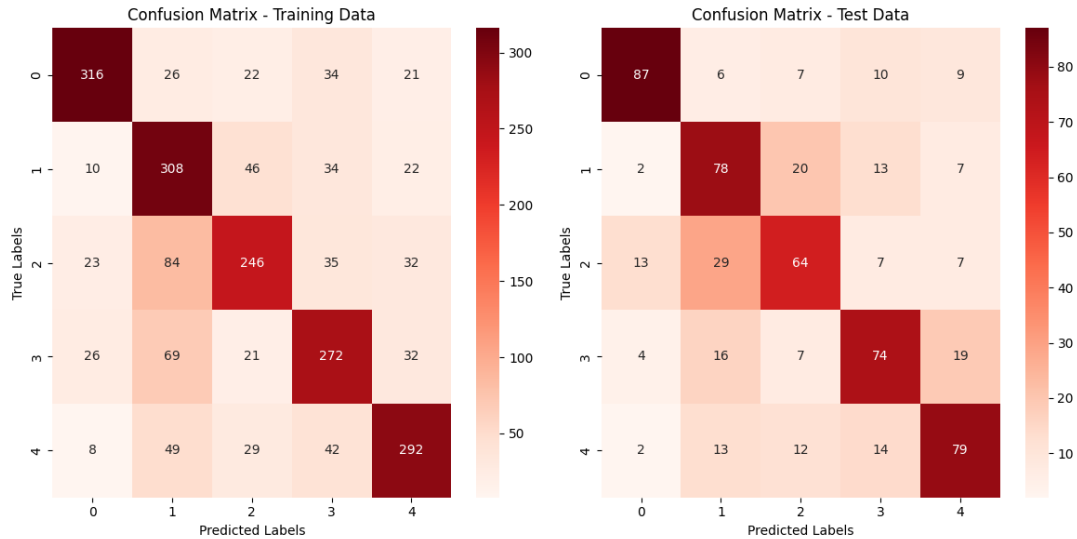


Figure 12: Confusion Matrices for GMM: Left - Training Data, Right - Test Data

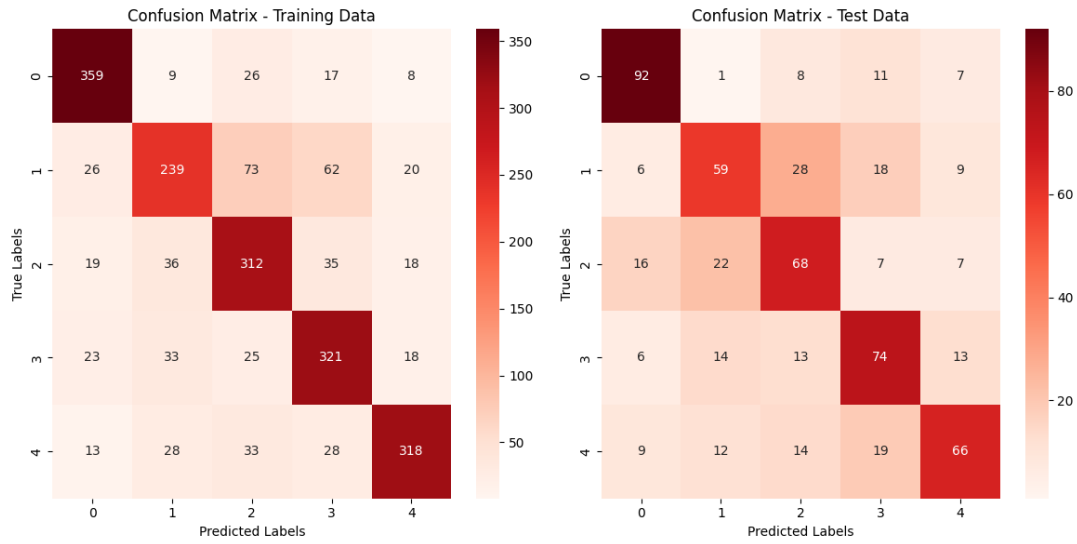


Figure 13: Confusion Matrices for MLFFNN: Left - Training Data, Right - Test Data

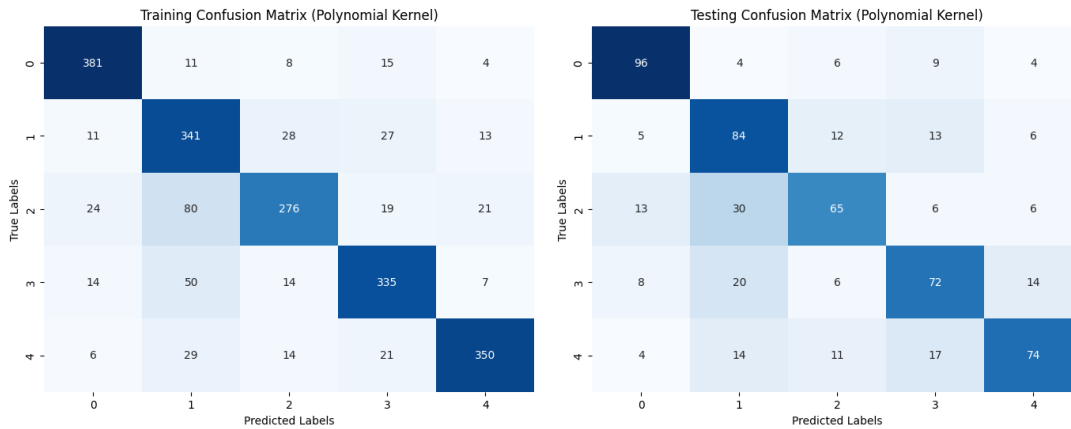


Figure 14: Confusion Matrices for SVM: Left - Training Data, Right - Test Data

5.3 Summary

In this exercise, Dataset 3 was classified using reduced-dimensional representations obtained from PCA. The SVM model based on the polynomial kernel (degree 2, $C = 1$) achieved the highest test accuracy (65.27%), outpacing the GMM and MLFFNN models. Confusion matrices indicate that while SVM performed effectively, there were still classification challenges present, particularly with certain classes. This analysis underscores the importance of dimensionality reduction for enhancing classifier performance, particularly in complex datasets like those encountered in this assignment.